First Steps

From the downloaded archive to the first box on screen, with no surprises in class

Level: all levels • **Time in class:** 20–30 min, once, before the first lesson • **Sensors:** all four

1. Installing the application

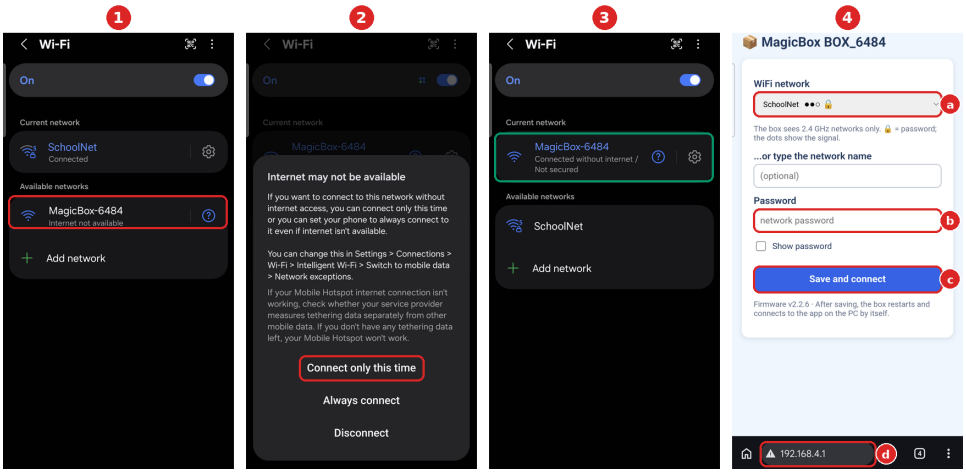
1. **Unzip the whole `MagicBox` folder** into *Documents* or onto a memory stick. **Not into `Program Files`**: Windows does not let the application write there, and the measurement logs then go to `Documents\MagicBox\logs`. Do not run the application from inside the ZIP archive, and do not move `MagicBox.exe` out of its folder.
2. Nothing else needs installing: start `MagicBox.exe`. If Windows shows the SmartScreen message *"Windows protected your PC"*, click **More info**, then **Run anyway**. The message appears because the application has no commercial signature, not because it is harmful.
3. On first start, **Windows Defender Firewall** asks whether the application may use the network. Tick **Private networks** and click **Allow access**. Without this step the boxes cannot connect over WiFi (the application announces itself to the boxes on UDP 8766 and receives them on TCP 8765). If the school network shows as *Public*, ask the IT technician to make it *Private* or to open those two ports.
4. The application starts **in English**, at the **Teacher** level, which shows every activity; for Romanian, French or German see below. The manuals (this document, the guides, the sheets) are in the `docs` folder next to the application; the **Open guide** button on each activity opens the right one.

2. Connecting a box over a USB cable

Sensors first. The box has **three sockets** for four sensors (light, temperature-humidity-pressure, CO_2, motion), so one is always left out: that is normal. With the box switched off, plug in the sensor the activity needs (the games and the tilt activities need the motion sensor); the box recognises it at power-on. The "sensor → activities" table is in the technical manual, chapter 2.

Connect the box with a USB Type-C *data* cable (some charging cables have no data wires). The **Home** page shows the box within a few seconds, with its name, colour and latest readings. If it does not appear within 10 seconds, click **Rescan**; if it still does not, Windows lacks the board's driver. Install it once, as administrator: **CP210x** (Silicon Labs), `https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers`, or **CH340** (WCH), `https://www.wch-ic.com/downloads/CH341SER_EXE.html`. In *Device Manager*, under *Ports (COM & LPT)*, the port's name tells you which one you need.

3. Connecting over WiFi



(1) tap **MagicBox-XXXX**; (2) *"Connect only this time"*; (3) connected, without internet: normal; (4) pick the network (a), type the password (b), *"Save and connect"* (c). If the page does not open by itself: `192.168.4.1` (d).

1. The first time, the box creates an open network **MagicBox-XXXX** (XXXX = the end of the box ID; **CutiaMagica-XXXX** before firmware 2.2.6). Join it from a phone and follow the picture. The box remembers the network and sees **only 2.4 GHz networks**: on a dual-band router, keep the 2.4 GHz band switched on.
2. The laptop and the boxes must be **on the same network**. The application finds them by itself; no address is ever typed. A box that fails to connect keeps retrying on its own, at least once a minute.

3. **School networks** with client isolation (devices cannot "see" each other) or with WPA2-Enterprise sign-in (a user name *and* a password) do not work. The simple fix: **a phone hotspot** or a small router just for the class; connect the laptop and the boxes to it.
4. To change the saved network, hold the **BOOT** button while powering the box.

4. The box firmware

There is one firmware, `serial_wifi`, current version **2.3.0**; it works over the USB cable as well as over WiFi. The application tells you itself when a box must be reflashed (firmware older than 2.2.1, or the old USB-only firmware) and when a newer version exists. Reflash following `firmware\FLASHING_GUIDE.pdf` in the package. The light sensor reaches direct sunlight only from 2.2.5; a box that reads about 3 800 lx in full sun has old firmware.

5. Settings to make once

All of them are under **Settings**, in the bar on the left.

- **Language and display**: under **Language:** choose Română, English, Français or Deutsch, then restart the application (the guides and manuals come in the same four languages). **Projector mode** makes chart text and lines larger for the big screen.
- **Box names**: every box that has connected at least once has a row. Give it a name ("Group 1", "Window"); the box's name and colour then appear on every screen and in exported files. Write the same name on the box. Here too: lay the box flat and look at acceleration Z on **Live Data**; it should be about +1 g. If it shows about −1 g, tick **Motion sensor upside down** for that box. From firmware 2.3.0, the tick is kept in the box.
- **Age level** (Junior, Explorer, Scientist, Teacher) ticks the recommended activities; the **Visible activities** list chooses exactly what the class sees.
- **Altitude Calibration** (optional): **Altitude of the school**, for the sea-level pressure in the Weather Station.

6. Where the data go

- Everything a box sends is written automatically to the `logs` folder next to the application (or to `Documents\MagicBox\logs`), one CSV file per box and session, plus an `_imu` one for motion. Nothing is ever deleted on its own.
- A guided experiment saves the pupils' journal there (`_journal.txt`) as they type, and at **Finish** an Excel file with the measurements of every step as well. The end screen says where.
- The **Data & Export** page (at the Scientist and Teacher levels) exports one box to Excel or CSV; **Export all boxes (Excel)** puts the whole lesson in one folder, one file per box. A CSV exported in Romanian opens in columns in Romanian Excel.

7. When something does not work

What you see	What to do
The activity says "No connected box has the … sensor plugged in"	The box lacks the sensor the activity needs: switch it off, plug the sensor in, switch it on. Which sensors a box has is written under **Settings** → **Advanced**, in the connection log ("Sensors it reports: …").
The box does not appear over USB	Another (data) cable; the CP210x/CH340 driver; close the Arduino IDE or any program holding the port; **Rescan**.
The box does not appear over WiFi	Laptop and box on the same network; the firewall permission (Private networks); a phone hotspot if the school network isolates clients.
The box appears, then vanishes	Weak power: another USB port or charger; weak WiFi: move the box closer to the router.
Light stops at ∼3 800 lx	Old firmware: reflash it (2.2.5 or newer).
A box flat on the table reads AZ ≈ −1 g	If the application asks for a reflash: firmware older than 2.2.1, reflash it. Otherwise the sensor is mounted the other way up: tick **Motion sensor upside down** for that box.
CO_2 is missing	The CO_2 sensor must be in a socket, and it needs about 30 s after power-on; a box with the old USB-only firmware: reflash it with `serial_wifi_v2_3_0`.
The Excel export is not saved	The file is open in Excel: close it and export again.
The logs are not next to the application	The application folder is read-only: look in `Documents\MagicBox\logs` or, better, move the application folder to *Documents*.
The application should be in Romanian	**Settings** → **Language:** Română, then restart.

Technical details: `docs\en\technical_manual.pdf`. What changed between versions: `CHANGELOG.txt` next to the application.